

Amanda Fonseca Rodrigues

amandafonsecarod09@gmail.com | +55 (88) 99486-5035 | Fortaleza/CE | LinkedIn | GitHub

Profile

Software Developer specializing in Frontend Engineering and Applied AI Systems. Experience in building production-ready interfaces with React and TypeScript, and implementing RAG (Retrieval-Augmented Generation) pipelines. Proven track record in leading development squads and designing scalable full-stack architectures. Focused on bridging the gap between complex AI backends and intuitive user experiences.

Experience

Atlântico – Employment Scholarship Fellow (Nexus AI)

2025 – Present

AI Engineering Fellow

- **Engineered** agent-based simulation personas (SCHUM & PETER) to automate complex innovation funding decision-making processes.
- **Optimized** RAG pipelines using Python, focusing on enhancing retrieval accuracy and LLM response reliability.
- **Implemented** advanced prompt engineering techniques and structured datasets to refine LLM behavioral consistency.
- **Collaborated** on the development of Nexus AI, bridging the gap between generative AI models and practical decision-making tools.

Cosmos Tech

2025 – Present

Frontend Developer

- **Architected** responsive and high-performance frontend features using React and TypeScript for AI-driven products.
- **Integrated** complex REST APIs with dynamic state management to ensure real-time AI data rendering.
- **Streamlined** deployment workflows using Vercel and maintained version control integrity through Git-based collaboration.

VORTEX Lab – PRISMA IAT Project (Unifor)

Feb 2025 – Sep 2025

Squad Leader – Volunteer

- **Led** a cross-functional development squad, overseeing the full lifecycle of a psychological research system (PRISMA IAT).
- **Orchestrated** technical alignment and delivery prioritization as the primary bridge between clients, tech leads, and academic advisors.
- **Ensured** project scalability by contributing to core frontend architecture and critical system feature implementations.

IFCE (Federal Institute of Ceará)

2022

Research Fellow – Machine Learning & IoT

- **Developed** intrusion detection systems for IoT environments utilizing Machine Learning models to identify network anomalies.
- **Analyzed** large datasets to evaluate model performance, focusing on precision and recall for anomaly detection.
- **Co-authored** a scientific publication detailing research methodology and experimental results in anomaly detection.

Selected Projects

Stock Portfolio System – Functional Programming Architecture

- **Full-stack Development:** Engineered a high-concurrency stock tracking system using **Clojure** and **ClojureScript** (Reagent/React).
- **Real-time Data:** Integrated Brapi API for live market data streaming and portfolio valuation logic.
- **Links:** Backend | Frontend.

CinemaGOF – Microservices Architecture (Spring Boot)

GitHub

- **Scalability:** Designed a distributed ticket management system using **Java** and **Spring Boot** microservices.
- **Service Mesh:** Implemented RESTful communication between modular services (Movie Catalog, Bookings).

VetCare – Veterinary Management System

GitHub

- **Full Architecture:** Developed the end-to-end system using **Node.js (Sequelize/MySQL)** and **React + TypeScript**.
- **Business Logic:** Built complex financial dashboards and appointment scheduling modules.

Other Relevant Projects:

- **Zeus (Full-Stack):** Pet expense tracker with React Native mobile app and Node.js/MongoDB backend.
- **UniQuadra:** Court reservation platform focused on responsive UI and complex booking logic.

Technical Skills

AI & Data: RAG (Retrieval-Augmented Generation), Prompt Engineering, LLM Evaluation, Python (Pandas/NumPy), LangChain concepts.

Frontend: React.js, TypeScript, JavaScript, React Native, Tailwind CSS, Responsive Design.

Backend: Node.js, Express.js, Java (Spring Boot), Clojure, RESTful APIs, Microservices.

Databases: MySQL, MongoDB, PostgreSQL, Vector Databases

Tools & DevOps: Git, Vercel, Docker, CI/CD, Agile Methodologies.

Education

University of Fortaleza (Unifor)

2023 – 2027

Bachelor's Degree in Computer Science

Fortaleza, CE

IFCE - Federal Institution of Education, Science and Technology

2022

Technical Degree in Computer Networks

Boa Viagem, CE

Languages

Portuguese (Native)

English (B2)